Indian Institute of Technology Jodhpur
2nd Year B.Tech Electrical Engineering — Curriculum Structure, August 2020

1. Introduction

The vision of the Department is to develop a world class teaching and research atmosphere by excelling in fundamental knowledge and applications of Electrical Engineering through curricular and co-curricular activities. The program provides strong fundamentals in mathematics and science which develop analytical thinking and form the basis of the diverse tool set used by electrical engineers to innovate, design and develop the technology of the future.

The curriculum offers flexibility to students for pursuing capability-linked specializations of their choice including: Cyber Physical Systems, Intelligent Communication and Networking, Communication Engineering, VLSI Systems, Visual Computing, Socio-Digital Reality, Nano and Flexible Electronics, Smart Grid, and Artificial Intelligence of Things.

2. Objectives of the Program

The program prepares graduates for three potential career paths:

1. An industrial path to take up the role of a technical leader or a manager through life-long learning.

2. An entrepreneurial path to apply the acquired knowledge to develop new products and initiate Deep Tech Start-up Company.

3. An academic path to enter a top-tier graduate program to conduct cutting-edge research and eventually pursue a research or academic profession.

3. Expected Graduate Attributes

1. Strong understanding of fundamentals of mathematics, science and engineering with emphasis on electrical technologies for Communication, Signal Processing, Devices & Circuits, Computing Systems, Control Engineering, and Power Engineering.

2. Ability to apply technical skills to identify, formulate, and solve complex problems encountered in modern electrical engineering practices.

3. Ability to model, analyze, design and experimentally evaluate electrical components or systems.

4. Ability to compete effectively in a world of rapid technological advancements and assume leadership roles.

5. Have established the foundations for critical thinking needed to broaden careers through minor/graduate-level studies.

6. Ability to recognize and practice professional ethics in working environment.

7. Ability to acquire new knowledge and integrate it to create new innovative electrical systems.

4. Learning Outcomes

1. Acquire knowledge of electrical engineering principles along with computing, engineering fundamentals, mathematics, and science.

2. Apply concepts of mathematics, sciences and engineering to analyze and design complex electrical and electronic circuits and systems.

3. Gain in-depth understanding of Signals, Devices, Circuits, Systems, Machine Learning, Programming, Control, Communications, Hardware Design.

4. Create, select, and apply appropriate techniques, resources, and computing tools to solve complex engineering problems.

5. Perform literature review and patent landscaping for innovative research.

6. Adapt to work in multidisciplinary environment.

7. Understand professional ethics and social responsibilities.

8. Develop technical presentation skills and communicate effectively.

9. Engage in independent learning in the context of technological advancements.

10. Develop attitude for product design and entrepreneurial activities.

5. New Skill Sets Targeted

- 5G and Beyond Communications
- Cyber Physical Systems
- Nano-electronics and Quantum Devices
- Smart Grid

6. Topic Clouds and Mapping with Courses


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[
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    "Area": "Communication",
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  {
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    "Topics": "Signal and System Classification, LTI Systems, Fourier, Laplace and z-Transform, Sampling",
    "Course": "Signals and Systems (IE, EE)"
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  {
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    "Course": "Engineering Electromagnetics (PC)"
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  {
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    "Topics": "Source Coding, Channel Coding, Channel Capacity Theorem, AWGN Channel Capacity, Multiuser Channe
    "Course": "Information Theory and Coding (PE)"
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    "Area": "Communication",
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    "Topics": "Sampling Theory and Signal Reconstruction, FFT, Digital Filters",
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    "Category": "Techniques",
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    "Category": "Techniques",
    "Topics": "Interfacing with GPIOs, Interfacing DC Motor, Keyboard Interfacing, Use of Timers, Implementatio
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    "Area": "Computing Systems",
    "Category": "Systems",
    "Topics": "Embedded Computing and SoC",
    "Course": "High Level Synthesis (PE, CS), Real-Time Systems (PE), Reconfigurable Computing (PE), Artificial
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"Category": "Techniques",
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"Category": "Systems",
"Topics": "Smart Grid",
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7. Course Categories and Credit Distribution (Regular B.Tech.)

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"Course_Type": "Institute Core (I)",
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"Course_Category": "Programme Compulsory (PC)",
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8. Credit Structure


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9. Programme Compulsory Courses


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  {
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    "LTP": "3-0-0",
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    "Credit": 3
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    "Contact_Hours": 3,
    "Credit": 3
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{
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  "Course_Name": "Analog Circuits",
  "LTP": "3-0-3",
  "Contact_Hours": 6,
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  "LTP": "3-0-0",
  "Contact_Hours": 3,
  "Credit": 3
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  "Contact_Hours": 3,
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  "LTP": "3-0-0",
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{
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  "LTP": "3-0-3",
  "Contact_Hours": 6,
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  "Contact_Hours": 3,
  "Credit": 2
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  "Sr_No": 12,
  "Course_Name": "Data Communication Networks",
  "LTP": "3-0-0",
  "Contact_Hours": 3,
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    "Contact_Hours": 3,
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10. Programme Elective Courses (Stream-wise)

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    "LTP": "3-0-0",
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    "Stream": "Communication",
    "Course": "Microwave Engineering (400)",
    "LTP": "3-0-0",
    "Credit": 3
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  {
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"Course": "Satellite Communications (400)",
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  "Course": "Biosensors (300)",
  "LTP": "3-0-2",
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  "LTP": "3-0-0",
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    "LTP": "3-0-0",
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11. Specializations

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"Elective_12cr": "Robust and optimal control, Nonlinear and Adaptive control, Modern Control, Networked Dyn
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"Elective_12cr": "Cellular Communication Networks, MIMO Wireless Communications, UAV Assisted Wireless Netw
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"Elective_12cr": "Embedded Systems, Microsystems Fabrication Technology, Sensors and Measurements, Deep Lea
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"Elective_12cr": "Digital and SoC Design, Fundamentals of Signal and Power Integrity, Image Sensor Design a
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12. Curriculum of B.Tech. EE (Regular) — Semester-wise

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Concentration Roadmaps (Tree Structure)

The following ASCII trees represent the course progression roadmaps for each EE concentration area.

EE UG - Communications Concentration Roadmap

```
First Year Courses
■■■ Pattern Recognition & Machine Learning
■   ■■■ (feeds into signal analysis)
■■■ Engineering Electromagnetics
■   ■■■ Digital Signal Processing
■       ■■■ Optical Communication Systems
■       ■■■ Microwave Engineering
```

```
■■■ Foundation of Quantum Information Processing
■   ■■■ (feeds into advanced topics)
■■■ Signal and Systems
■   ■■■ Communication Systems
■       ■■■ Information Theory and Coding
■       ■   ■■■ Optical Communication Systems
■       ■   ■■■ Cellular Communication Networks
■       ■   ■■■ Spread Spectrum Communications
■       ■   ■■■ Satellite Communications
■       ■■■ Data Communication Networks
■■■ Probability, Statistics and Stochastic Processes
    ■■■ Information Theory and Coding
```

EE UG - Signal Processing Concentration Roadmap

```
First Year Courses
■■■ Pattern Recognition and Machine Learning
■   ■■■ (input to signal interpretation)
■■■ Probability, Statistics and Stochastic Processes
■   ■■■ Digital Signal Processing
■       ■■■ Signal Compression
■       ■   (Lossless compression, Sparse Representation, JPEG, MPEG)
■       ■■■ Speech Processing
■       ■   (Time/Frequency Domain, Feature Analysis, Music Classification)
■       ■■■ System Identification
■       ■   (Nonparametric, Parametric, Echo/Noise Cancellation, Deconvolution)
■       ■■■ Digital Image Processing
■           (Image Formation, Transform Theory, Filtering, Segmentation)
■■■ Data Structure and Algorithms
■   ■■■ (algorithmic foundations for processing)
■■■ Signal and Systems
    ■■■ Information Theory and Coding
```

EE UG - Control Engineering Concentration Roadmap

```
First Year Courses
■■■ Circuit Theory
■   ■■■ Control Systems
■       ■■■ Modern Control
■           ■■■ Nonlinear and Adaptive Control
■           ■   (Limit Cycles, Adaptive Systems, Reinforcement Learning based Control)
■           ■■■ Foundations of Cyber-Physical Systems
■           ■   (CPS Components, Principles of Dynamical Systems, CPS Implementation)
■           ■■■ Robust and Optimal Control
■           ■   (Stability and Optimal Control Theory, Robust Control for Nonlinear Systems)
■           ■■■ Introduction to Robotics
■               (Robot Modelling, Motion Planning, Vision based Control, Robot Control)
■■■ Signal and Systems
■   ■■■ (feeds into Control Systems)
■■■ Probability, Statistics and Stochastic Processes
    ■■■ (feeds into Modern Control)
```

EE UG - Devices and Circuits Concentration Roadmap

```
First Year Courses
■■■ Foundation of Quantum Information Processing
■   ■■■ VLSI Design
■       ■■■ CAD for VLSI
■       ■   (VLSI Design Automation, Algorithmic Techniques, Optimization based Analog/RF Design)
■       ■■■ Physics and Modeling of MOS Transistor
■       ■   (MOS Transistor: Building block of IC, operation and modeling)
■       ■■■ IC Technology
■       ■   (Steps in IC technology, Fabrication Processes)
■       ■■■ Sensors and Actuators
■           (Basic Principles, Characterization, Read out circuits)
■■■ Thermodynamics → Material Science and Engineering → Physical Electronics → Analog Circuits
■■■ Circuit Theory → Physical Electronics → Analog Circuits
```

■■■ Signal and Systems

EE UG - Computing Systems Concentration Roadmap

First Year Courses
■■■ Digital Systems Design
■ ■■■ Digital Systems Lab
■ ■■■ (feeds into embedded and advanced computing)
■■■ Pattern Recognition and Machine Learning
■ ■■■ (ML context for computing systems)
■■■ Signal and Systems
■ ■■■ Digital Signal Processing
■ ■■■ (signal computation background)
■■■ Data Structure and Algorithms
■ ■■■ Computer Architecture
■ ■■■ Embedded Systems
■ ■■■ High Level Synthesis
■ ■ (VLSI Design Flow, Behavioural synthesis, Synthesis Tools, Popular chip architectures)
■ ■■■ Reconfigurable Computing
■ ■ (Implementation architectures, Domain specific processors, FPGA/ASIC-based accelerators)
■ ■■■ Real-Time Systems
■ (Real-time scheduling Algorithms, Bandwidth Servers, Real-Time Operating Systems)
■■■ (all feed into Digital Systems Lab)

EE UG - Power Engineering Concentration Roadmap

First Year Courses
■■■ Signals and Systems → (background for analysis)
■■■ Material Science and Engineering → Physical Electronics → Electrical Machines (EM) and EM Lab
■■■ Physical Electronics → Power Engineering
■■■ Engineering Electromagnetics → Power Engineering
■■■ Circuit Theory → Electrical Machines (EM) and EM Lab → Power Engineering
■■■ Power System Analysis and Stability
■ (Bus admittance matrix, Load flow analysis, Fault analysis, Power system stability/protection/transient
■■■ Power System Operation and Control
■ (Power generating units, Economic dispatch, Unit commitment, Power system security, Load frequency cont
■■■ Power Electronics
■ (Power electronic switches, Controlled rectifiers, DC-DC converters, Inverters, AC voltage controllers)
■■■ Industrial Drives
(Starting/braking/speed control, 4 quadrant drives, DC motor drives, 3-phase induction/synchronous moto